MILITARY SPECIFICATION

INDICATOR, ATTITUDE, TYPE J-8

This amendment forms a part of MIL-I-5133C(USAF), dated 5 Aug 68, and is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

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* Delete "2.1" and add:
"2.1 Government documents."

* Add:
"2.1.1 Specifications and standards. The following specifications and standard form a part of this specification to the extent specified herein. Unless other specified, the issues of these documents shall be those listed in the Department Defense of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation."

* Under SPECIFICATIONS, Federal, delete: "QQ-P-416 Plating, Cadmium (Electrodeposited)". Add: "PPP-B-601, Boxes, Wood, Cleated-Plywood" and "PPP-B-636 Boxes, Shipping, Fiberboard".

* Under SPECIFICATIONS, Military, delete "MIL-E-5272" and add: "MIL-C-83488 Coatin Aluminum, Ion Vapor Deposited".

Add boxed paragraph to bottom center of page.

Beneficial comments (recommendations, additions, deletions) and any pertinent dat which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1 Under STANDARDS, Military, delete: "MIL-STD-834 Packaging Data Forms,
* Instructions For Preparation And Use Of" and add: "MIL-STD-810C Environmental Te Methods".

* Add:
"2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the

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issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM-D-3951-82 Packaging, Commercial - 29 Jan 82

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)"

3.1 Delete and add:
"3.1 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.2.a and 6.4)."

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3.9.2 Delete and add:
"3.9.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

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4.2.a, Delete and add:
"4.2.a First article inspection (see 4.3)."

4.3 Delete "Qualification" two places and add "First Article".

* Paragraph 4.3, under "Qualification Inspection, item h": Delete and add: "Salt Fog".

4.3.1 Delete and add:
"4.3.1 First article test sample. The first article test sample shall consist of five indicators representative of the production equipment. These indicators shal be identified as first article samples and forwarded to the contracting activity."

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4.4.2.h, Delete.

* Paragraph 4.6.15: Delete and add: "The indicator shall be properly connected, mounted and subjected to a vibration test in accordance with Method 514.2, Procedure I, Part I, Table 514.II, Figure 514.2-2, Curve C, Category b.1 of MIL-STD-810C. The cycling time shall be two complete sweeps in each axis. For th test, only the sinusoidal cycling portion will be required and Curve C shall be modified as follows: A range of 5 to 50Hz where a displacement of 0.1 inch double amplitude from 5 to 23 Hz and ±1g from 23 to 50Hz. The reading during testing shall not differ by more than 1 degree from reading obtained at rest."

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* Paragraph 4.6.16., First Sentence: delete and add: "The low temperature exposure test shall be conducted in accordance with Method 502.1, Procedure I of MIL-STD-810C".

* Pargraph 4.6.17, First Sentence: Delete and add: "The high temperature exposure test shall be conducted in accordance with Method 501.1, Procedure I of MIL-STD-810C, except for a period of 24 hours."

* Paragraph 4.6.18, First Sentence: Delete and add: "The altitude test shall be conducted in accordance with Method 500.1, Procedure 1 of MIL-STD-810C for a perio of 4 hours and at a chamber temperature of -54°C."

* Paragraph 4.6.19, Delete and add: "The rain test shall be conducted in accordance with Method 506.1, Procedure II of MIL-STD-810C. The indicator should not be operating during exposure."

* Paragraph 4.6.20, Delete and add: "The indicator shall be subjected to a acceleration test in accordance with method 513.2, Procedure I of MIL-STD-810C. A acceleration level of 20g shall be applied for a period of 1.0 minute along each o its three axes, in both directions while not operating. At the end of acceleration, the indicator shall be subjected to and shall pass the tests specified under 4.4.1. No damage shall result from this test."

* Paragraph 4.6.21, Delete and add: "The indicator shall be subjected to a vibratio test over the frequency range from 5 to 50Hz in accordance with Method 514.2, Procedure I, Part 1, Figure 514.2-2 of MIL-STD-810C. The indicator shall be teste over the frequency range from 5 to 33Hz of Curve C and 33 to 50Hz of Curve D. At the conclusion of this test, the indicator shall be subjected to the roll, pitch and yaw test. No looseness in the mechanism nor damage to any part of the indicator shall result from the vibration."

* Paragraph 4.6.22, First Sentence: Delete and add: "The sand and dust test shall be conducted in accordance with Method 510.1, Procedure I of MIL-STD-810C. At en of this test, the indicator shall then satisfactorily meet the test requirements specified for starting and alignment and roll, pitch and yaw."

* Paragrpah 4.6.23, Test Title and First Sentence: Delete and add: "Salt fog". T salt fog test shall be conducted in accordance with Method 509.1, Procedure I of MIL-STD-810C for a period of 50 hours with the indicator not operating.

* Paragraph 4.6.24, First Sentence: Delete and add: "The humidity test shall be i accordance with Method 507.1, Procedure I of MIL-STD-810C with the indicator not operating."

* Paragraph 4.6.25, First Sentence: Delete and add: "The fungus resistance test shall be conducted in accordance with Method 508.1 of MIL-STD-810C."

4.6.29, Delete.

* Add new paragraphs 4.7 thru 4.7.2:

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"4.7 Packaging inspection.

4.7.1 Quality conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with Group A and B quality conformance inspection requirements, Section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

4.7.2 First article. When the unit container is capable of serving as the shipping container First Article Inspection, and rough handling tests as outlined in Section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction."

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Delete present section 5 and add:
"5 PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C as specified by contracting activity.

5.1.1 Level A.

5.1.1.1 Cleaning. Unit shall be cleaned in accordance with process C-1 of MIl-P-116.

5.1.1.2 Drying. Immediately after cleaning, the unit shall be dried following ar one or combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

5.1.1.3 Preservation application. Not required.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each unit shall be packaged in quantity unit packs of one each in accordance with method 1C-1 of MIL-P-116. Overbox each completed pack in a PPP-B-636 fiberboard, weather-resistant class box. Cushioning material between bag and container shall be of type, density, and thickness that will protect the unit from impact shock damage when tested as specified in paragraph 4.7.2 of this specification.

5.1.2 Level C. Units shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity

* "5.1.3 Industrial. The industrial preservation of indicators shall be in accordance with ASTM-D-3951-82."

* "5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2)."

5.2.1 Level A. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636 weather-resistant. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636.

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For the Army, specification PPP-B-601 plywood box, shall be used for the shipping container. Insofar as practicable, exterior containers shall be of a uniform sha and size and of minumim cube and tare consistent with the protection required.

5.2.2 Level B. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636 class domestic. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, the shipping containers shall conform to specification PPP-B-636 class weather-resistant.

5.2.3 Level C. Packaged as specified in 5.1 shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

* "5.2.4 Industrial. The packaged indicators shall be packed in accordance with ASTM-D-3851-82."

5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance witl MIL-STD-129."

6.4 Delete and add:
"6.4 First article. A first article is required for inspection and approval (see 3.1 and 4.2.a)."

6.6 Add:
"6.6 Reclaimed materials. The use reclaimed materials shall be encouraged to the maximum extent possible."

The margins of this amendment are marked with an asterisk to indicate where change (additions, modifications, corrections, deletions) from the previous amendment wer made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire conter irrespective of the marginal notations and relationship to the last previous amendment.

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